Adaptive Bandwidth, Score Homogeneity, and Conditional Coverage in CKME-CP

Abstract

We identify a new regime for conformal prediction with CDF-based scores: when the bandwidth of the smooth indicator function is proportional to the local noise scale, the conformal score distribution becomes approximately x-independent, even though the underlying CDF estimate is biased. This "score homogeneity" property ensures that a single calibration threshold provides uniform conditional coverage — a guarantee unavailable to fixed-bandwidth estimators or quantile-first methods. We formalize the mechanism through the concept of *effective bandwidth in standardized scale*, prove an over-smoothed collapse proposition showing what fails when homogeneity is achieved trivially, and validate the theory with a three-phase experiment covering mild, extreme, and non-Gaussian heteroscedasticity.

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1 Setup and Notation

Data model. $Y = f(x) + \sigma(x)\varepsilon$, where $f(x) = E[Y \mid X = x]$, $\sigma(x) > 0$ is the conditional noise scale, and $\varepsilon \sim G(\cdot)$ with $E[\varepsilon] = 0$, $\mathrm{Var}(\varepsilon) = 1$, independent of x.

CKME CDF estimator. Given training data (X_i, Y_{ij}) $(i = 1, \ldots, n;\ j = 1, \ldots, r)$, CKME estimates:

$$\hat{F}(y \mid x) = \sum_{i=1}^{n} c_i(x) \left[\frac{1}{r} \sum_{j=1}^{r} \Psi\left(\frac{y - Y_{ij}}{h} \right) \right], \tag{1}$$

where $\Psi(u) = 1/(1 + e^{-u})$ is the logistic CDF, $h > 0$ is the bandwidth, and $c_i(x)$ are kernel regression weights satisfying $\sum_i c_i(x) \approx 1$.

Conformal score. The abs-median score is:

$$s(x, y) = |\hat{F}(y \mid x) - 0.5|. \tag{2}$$

Standardized coordinates. Define $z = (y - f(x))/\sigma(x)$ and $Z_{ij} = (Y_{ij} - f(x))/\sigma(x)$.

2 Adaptive Bandwidth and Score Homogeneity

2.1 The Effective Bandwidth

The smooth indicator in (1) evaluates:

$$\Psi\left(\frac{y - Y_{ij}}{h} \right) = \Psi\left(\frac{(y - f(x)) - (Y_{ij} - f(x))}{h} \right) = \Psi\left(\frac{\sigma(x)(z - Z_{ij})}{h} \right) = \Psi\left(\frac{z - Z_{ij}}{h/\sigma(x)} \right).$$

Definition 1 (Effective bandwidth)**.** *The* ***effective bandwidth in standardized scale*** *is*

$$h_{\mathrm{eff}}(x) \ = \ \frac{h}{\sigma(x)}. \tag{3}$$

It controls the smoothing applied to the standardized noise CDF $G(z)$: a larger h_{eff} produces a smoother (more biased) estimate.

Intuition. The bandwidth h operates in the original y-scale, but the "natural unit" for smoothing is the noise scale $\sigma(x)$. When h is fixed, the estimator over-smooths where noise is small and under-smooths where noise is large.

2.2 Fixed vs. Adaptive Bandwidth

Proposition 2 (Score homogeneity under adaptive bandwidth)**.** *Under the location-scale model $Y = f(x) + \sigma(x)\varepsilon$ with $\varepsilon \sim G(\cdot)$ independent of x:*

(a) ***Fixed*** *h: $h_{\mathrm{eff}}(x) = h/\sigma(x)$ varies with x. The CDF estimate in standardized scale is $\hat{G}_{h/\sigma(x)}(z)$, which depends on x through $\sigma(x)$. The score distribution $F_{s|x}$ is x-dependent.*

(b) ***Adaptive*** $h(x) = c\,\sigma(x)$***:*** $h_{\text{eff}}(x) = c$ *(constant). The CDF estimate becomes*

$$\hat{F}(y \mid x) = \sum_i c_i(x) \left[\frac{1}{r} \sum_j \Psi\left(\frac{z - Z_{ij}}{c} \right) \right] \approx \hat{G}_c(z), \tag{4}$$

a common kernel-smoothed estimate of $G(z)$ with bandwidth c, independent of x. Consequently, the score $s(x, Y) = |\hat{G}_c(Z) - 0.5|$ has an x-independent distribution.

Intuition. Adaptive bandwidth "standardizes" the smoothing: every x sees the same effective resolution c when measured in units of the local noise. The CDF bias $\hat{G}_c(z) - G(z)$ is a function of c and G only, not of x.

Proof sketch. Under $h(x) = c\,\sigma(x)$, the argument of Ψ becomes $(z - Z_{ij})/c$ as shown above. Since $Z_{ij} \sim G(\cdot)$ independently of x, the site-averaged indicator $\frac{1}{r}\sum_j \Psi((z - Z_{ij})/c)$ is a kernel density estimate of G at z with bandwidth c, using i.i.d. samples from G. The dependence on x enters only through $c_i(x)$, which are kernel regression weights satisfying $\sum_i c_i(x) \approx 1$. Under standard kernel regression consistency conditions, $\hat{F}(y|x) \to \hat{G}_c(z)$ as $n \to \infty$, where $\hat{G}_c$ is the pooled kernel-smoothed CDF of G. □

Remark 3 (Why CDF-first is essential). Quantile-first scores such as $s(x, y) = \max(\hat{q}_{\text{lo}}(x) - y, y - \hat{q}_{\text{hi}}(x))$ do not have PIT structure. Even if the quantile estimator has x-homogeneous bias, the score distribution depends on the local noise scale through the y-space comparison. CDF-first scores, by operating in probability space $[0, 1]$, automatically inherit the standardization that makes homogeneity possible.

2.3 Over-Smoothed Collapse

Score homogeneity alone is insufficient: trivially setting $h \to \infty$ also makes $\hat{F}(y|x)$ independent of x (all CDFs flatten to 1/2), but the resulting CP intervals are useless.

Proposition 4 (Over-smoothed collapse)**.** *In the regime $h \to \infty$ (equivalently, fixed h with $\sigma(x) \ll h$ for all x), the abs-median CP interval collapses to a global constant width:*

$$W(x) = U(x) - L(x) \;\to\; 2\,Q_{1-\alpha}\big(\{|Y_j - \hat{f}(X_j)|\}_{j=1}^{n_{\text{cal}}}\big) \quad \textit{for all } x, \tag{5}$$

where $Q_{1-\alpha}$ denotes the $(1-\alpha)$-quantile of the calibration residuals.

Proof sketch. The logistic CDF admits the Taylor expansion $\Psi(u) \approx \frac{1}{2} + \frac{u}{4}$ for $|u| \ll 1$. When h is large, $u = (y - Y_{ij})/h$ is small, so:

$$\hat{F}(y \mid x) \approx \sum_i c_i(x) \left[\frac{1}{2} + \frac{y - \bar{Y}_i}{4h} \right] = \frac{1}{2} + \frac{y - \hat{f}(x)}{4h},$$

where $\hat{f}(x) = \sum_i c_i(x)\bar{Y}_i$ is the kernel regression estimate. The abs-median score becomes:

$$s(x, y) = \left| \hat{F}(y|x) - \frac{1}{2} \right| = \frac{|y - \hat{f}(x)|}{4h}.$$

The CP threshold $\hat{q}$ is the $(1-\alpha)$-quantile of $\{|Y_j - \hat{f}(X_j)|/(4h)\}$. The prediction interval $\{y : s(x, y) \le \hat{q}\}$ requires $|y - \hat{f}(x)|/(4h) \le \hat{q}$, giving:

$$L(x) = \hat{f}(x) - 4h\hat{q}, \qquad U(x) = \hat{f}(x) + 4h\hat{q}.$$

The width $U(x) - L(x) = 8h\hat{q}$. But $\hat{q} = Q_{1-\alpha}(|Y_j - \hat{f}(X_j)|)/(4h)$, so $8h\hat{q} = 2\,Q_{1-\alpha}(|Y_j - \hat{f}(X_j)|)$. The factor h cancels exactly, and the width depends only on the marginal distribution of calibration residuals. □

Remark 5 (Width floor and coverage distortion). The collapsed width $W^* = 2\,Q_{1-\alpha}(|\sigma(X)\varepsilon|)$ depends on the *marginal* distribution of $\sigma(X)$, not the local $\sigma(x)$:

- Where $\sigma(x) \ll E[\sigma(X)]$: the interval is vastly too wide → severe over-coverage (wasted precision).
- Where $\sigma(x) \gg E[\sigma(X)]$: the interval is too narrow → under-coverage.

Remark 6 (Numerical example: `exp2`). For `exp2`: $\sigma(x) = \sqrt{0.01 + 0.2(x-\pi)^2}$, $x \in [0, 2\pi]$, $\alpha = 0.1$.

- At $x = \pi$: $\sigma(\pi) = 0.1$, oracle 90% interval width $\approx 2 \times 1.645 \times 0.1 = 0.33$.
- At $x = 0$: $\sigma(0) \approx 2.0$, oracle width ≈ 6.53.
- Collapsed width: $W^* = 2\,Q_{0.9}(|\sigma(X)\varepsilon|) \approx 2.95$ (computed from the marginal residual distribution with $X \sim \text{Uniform}[0, 2\pi]$).
- At $x = \pi$: $W^*/W_{\text{oracle}} \approx 9.0$ (9× too wide).
- At $x = 0$: $W^*/W_{\text{oracle}} \approx 0.45$ (55% under-coverage risk).

2.4 The Two Conditions for Useful Homogeneity

The over-smoothed collapse shows that score homogeneity is *necessary but not sufficient*. Two conditions are needed simultaneously:

1. **Score homogeneity:** $d_{\text{TV}}(F_{s|x}, F_{s|x'}) \approx 0$ for all x, x'. This ensures a single $\hat{q}$ works uniformly.
2. **Local adaptivity:** The interval width $W(x)$ reflects local $\sigma(x)$. This ensures the interval is not trivially constant.

Adaptive bandwidth $h(x) = c\,\sigma(x)$ with moderate c achieves both: the CDF is smoothed uniformly in standardized scale (condition 1), but the smoothing preserves the $\sigma(x)$-dependence of the quantiles because the CDF estimate still distinguishes different $\sigma(x)$ through the y-to-z transformation (condition 2).

Fixed h with $h \to \infty$ achieves condition 1 only, violating condition 2 (Proposition 4). Fixed h with $h \to 0$ achieves condition 2 only (sharp CDF, locally adaptive width), but the score distribution is x-dependent, violating condition 1.

2.5 Connection to CP^2 Framework

Our score homogeneity result connects to the conditional conformal prediction literature through the total variation framework [3]. Under approximate exchangeability, the conditional coverage satisfies

$$P(Y \in C(x) \mid X = x) \geq 1 - \alpha - d_{\text{TV}}(x) - p_{n+1}(x), \tag{6}$$

where $d_{\text{TV}}(x)$ measures the discrepancy between the conditional and marginal score distributions and $p_{n+1}(x)$ is an exchangeability correction. The CP^2 framework [1] provides RKHS-based bounds on a similar coverage gap.

When $d_{\mathrm{TV}}(x) = \delta$ (constant). The bound becomes $P(Y \in C(x) \mid X = x) \geq 1 - \alpha - \delta - p_{n+1}$, a *uniform* bound independent of x.

Our contribution. CP^2 does not identify when $d_{\mathrm{TV}}(x)$ is x-independent. We provide a concrete sufficient condition:

Lemma 7 (Adaptive h implies uniform d_{TV})**.** *Under the location-scale model with $h(x) = c\,\sigma(x)$, the CDF-based conformal score distribution satisfies $d_{\mathrm{TV}}(F_{s|x}, F_{s|x'}) \to 0$ for all x, x' as $n \to \infty$. Hence $d_{\mathrm{TV}}(x)$ (relative to the pooled score distribution) is asymptotically constant in x.*

Important subtlety. Uniform d_{TV} alone is not sufficient (Section 2.3). The combination of small constant d_{TV} *and* locally adaptive intervals requires the bandwidth to be proportional to $\sigma(x)$, not merely large.

3 Coverage Bound and Rate Analysis

RCP (Rectified Conformal Prediction) [2] and our CKME-based approach both aim to reduce the x-dependence of conformal score distributions, but they operate on fundamentally different error decompositions. We make this distinction precise.

RCP's error structure. RCP's strategy is to transform the conformal score via a rectification map $s \mapsto \tilde{s} = T(s, x)$ so that the transformed score's conditional α-quantile $\tau(x) := Q_\alpha(\tilde{s} \mid X = x)$ becomes x-independent. In practice, $\tau(x)$ is unknown and must be estimated. Let $\hat{\tau}(x)$ denote the estimated conditional quantile (obtained, e.g., via local quantile regression with kernel bandwidth b). RCP's conditional coverage theorem [2] shows that the coverage gap is controlled by

$$\big|P(Y \in C(x) \mid X = x) - (1 - \alpha)\big| \lesssim \sup_x |\hat{\tau}(x) - \tau(x)|, \tag{7}$$

i.e., the *quantile-of-score estimation error*. This error in turn inherits the usual bias–variance trade-off of the local quantile regression bandwidth b: small b yields low bias but high variance; large b reverses the trade-off. The bandwidth b is a nuisance parameter that must be tuned, and its optimal value depends on the smoothness of $\tau(x)$.

Our error structure. In the CKME framework with CDF-based scores, we do not estimate a conditional quantile of the score. Instead, the relevant error decomposition is:

$$d_{\mathrm{TV}}\big(F_{s|x},\ F_{s,\mathrm{pivot}}\big) \ \leq\ \underbrace{\frac{\pi^2 c^2}{6}\,\|g''\|_1}_{\text{(i) smoothing bias}} \ +\ \underbrace{\sup_y |\hat{F}(y \mid x) - F(y \mid x)|}_{\text{(ii) CDF estimation error}}, \tag{8}$$

where term (i) arises from the logistic smooth indicator with adaptive bandwidth $h(x) = c\,\sigma(x)$ (Devroye and Györfi [4]), and term (ii) is the CKME regression error (which is $O_p(n^{-s})$ for some rate $s > 0$ depending on kernel smoothness and dimension).

When $\sigma(x)$ is unknown and replaced by $\hat{\sigma}(x)$, a third term appears:

$$\text{(iii) plug-in error:} \quad |\hat{h}(x) - h(x)| = c\,|\hat{\sigma}(x) - \sigma(x)|, \tag{9}$$

which propagates into both terms (i) and (ii) through the bandwidth. Bounding this term quantitatively is addressed in Section 3.1.

Structural comparison. The key difference is summarized as follows:

	RCP	**CKME (ours)**
Core estimated object	$\hat{\tau}(x)$: conditional quantile of score	$\hat{F}(y \mid x)$: conditional CDF
Coverage-controlling error	$\sup_x \lvert\hat{\tau}(x) - \tau(x)\rvert$	$d_{\mathrm{TV}}(F_{s\mid x}, F_{s,\mathrm{pivot}})$
Bandwidth role	nuisance parameter in local QR	design parameter with explicit bias formula
Error decomposition	QR bias–variance (implicit in b)	smoothing bias $O(c^2)$ + CDF error $O_p(n^{-s})$
Plug-in parameter	$\tau(x)$ itself	$\sigma(x)$ (noise scale)

Remark 8 (Different "quantile errors"). Both frameworks involve quantile estimation, but the quantile objects are distinct. RCP estimates $Q_\alpha(\tilde{s} \mid X = x)$, the α-quantile of the *transformed score* conditional on x. CKME, when used for prediction intervals, estimates $Q_\tau(Y \mid X = x)$, the τ-quantile of the *response* conditional on x, obtained by inverting $\hat{F}(y \mid x)$. The former is an intermediate quantity for calibration; the latter is the final inferential target. This distinction is important: our quantile error enters through CDF inversion and is bounded by the CDF estimation error via $|Q_\tau - \hat{Q}_\tau| \leq \|\hat{F} - F\|_\infty / f(Q_\tau \mid x)$, whereas RCP's quantile error is a direct estimation problem in its own right.

Implications for bandwidth selection. In RCP, the bandwidth b of the local quantile regression is a nuisance parameter: it controls the estimation quality of $\hat{\tau}(x)$ but has no structural role in the score distribution itself. In our framework, the bandwidth c of the smooth indicator plays a dual role: it is both an estimation parameter (controlling CDF smoothing bias) and a *design* parameter (controlling score homogeneity). The explicit decomposition (8) makes this dual role transparent and provides a principled basis for selecting c, which has no direct analogue in the RCP framework.

Two-sided conditional coverage bound. A structural advantage of the d_{TV} framework is that it yields a *two-sided* bound on conditional coverage. Since total variation controls discrepancy in both directions, we obtain:

Theorem 9 (Two-sided conditional coverage bound)**.** *Let $\hat{q}$ be the split-conformal calibration quantile computed from n calibration scores. Suppose $d_{\mathrm{TV}}(F_{s|x}, F_{s,\mathrm{pool}}) \leq \delta(c, n)$ for all x, where $F_{s,\mathrm{pool}}$ is the marginal (pooled) score distribution. Then*

$$\big|P(Y \in C(x) \mid X = x) - (1 - \alpha)\big| \;\leq\; \delta(c, n) + p_{n+1}, \tag{10}$$

where $p_{n+1} = O(1/n)$ is the finite-sample calibration correction. Under the adaptive bandwidth $h(x) = c\,\sigma(x)$ with the location-scale model, the bound (8) *gives*

$$\delta(c, n) \;\leq\; \frac{\pi^2 c^2}{6}\,\|g''\|_1 + O_p(n^{-s}).$$

Proof sketch. By definition of d_{TV}, $|P(s \leq \hat{q} \mid X = x) - P(s \leq \hat{q})| \leq d_{\mathrm{TV}}(F_{s|x}, F_{s,\mathrm{pool}})$. Standard split-conformal calibration gives $|P(s \leq \hat{q}) - (1 - \alpha)| \leq p_{n+1}$. The triangle inequality yields (10). The bound on $\delta(c, n)$ follows from (8). □

Remark 10 (Two-sided vs. one-sided bounds). RCP's coverage theorem is essentially one-sided: it guarantees that conditional coverage does not fall too far below $1 - \alpha$ (a safety guarantee), but does not directly control over-coverage. Proposition 9 controls both directions:

- The **lower bound** ensures safety (no severe under-coverage at any x).
- The **upper bound** ensures efficiency (no severe over-coverage, which would imply unnecessarily wide intervals at some x).

Together, they assert that conditional coverage is approximately $1-\alpha$ *uniformly* over x — a near-conditional-validity statement. This two-sided nature is a direct consequence of using d_{TV} (an absolute discrepancy) rather than a one-sided quantile perturbation argument.

The absolute-value form (10) also corresponds directly to the experimental metric `cov_gap_sup` $= \sup_x |\mathrm{coverage}(x) - (1-\alpha)|$ reported in the empirical results, making the theory–experiment connection transparent.

3.1 Plug-in Bandwidth for the Noise Scale

The theoretical adaptive rule $h(x) = c\,\sigma(x)$ requires the unknown noise scale $\sigma(x)$. In practice we replace it by a plug-in estimator $\hat{\sigma}(x)$, which introduces a third error component, $|\hat{h}(x) - h(x)| = c\,|\hat{\sigma}(x) - \sigma(x)|$, that was left unquantified in (9). Here we derive its rate when $\hat{\sigma}$ is built from replicated Stage-1 data.

Two-step construction of $\hat{\sigma}$. Assume the Stage-1 budget consists of n_0 sites $X_1, \ldots, X_{n_0} \sim p_X$, each with $r_0 \geq 2$ replications $Y_{i,1}, \ldots, Y_{i,r_0}$. Form the site-wise sample standard deviation

$$\tilde{\sigma}_i \;=\; \sqrt{\tfrac{1}{r_0-1}\sum_{j=1}^{r_0}(Y_{i,j} - \bar{Y}_i)^2}, \qquad \bar{Y}_i = \tfrac{1}{r_0}\sum_j Y_{i,j},$$

and smooth across X via a Nadaraya–Watson estimator with kernel K and bandwidth h_σ:

$$\hat{\sigma}_{h_\sigma}(x) \;=\; \frac{\sum_{i=1}^{n_0} K_{h_\sigma}(x - X_i)\,\tilde{\sigma}_i}{\sum_{i=1}^{n_0} K_{h_\sigma}(x - X_i)}. \tag{11}$$

Under $Y = f(x) + \sigma(x)\varepsilon$ with $\mathbb{E}[\varepsilon] = 0$, $\mathrm{Var}(\varepsilon) = 1$, a standard delta-method expansion gives

$$\mathbb{E}[\tilde{\sigma}_i \mid X_i] = \sigma(X_i) + O(r_0^{-1}), \quad \mathrm{Var}(\tilde{\sigma}_i \mid X_i) = \frac{\sigma^2(X_i)\,(\kappa_4 - 1)}{4(r_0 - 1)} + O(r_0^{-2}),$$

where $\kappa_4 = \mathbb{E}[\varepsilon^4]$.

Bias–variance decomposition. Under the usual NW assumptions (twice-differentiable σ and p_X, $p_X(x) > 0$, symmetric kernel K with $\mu_2(K) = \int u^2 K(u)\,du$, $R(K) = \int K^2$), the pointwise bias and variance of $\hat{\sigma}_{h_\sigma}$ satisfy, as $n_0 \to \infty$, $h_\sigma \to 0$, $n_0 h_\sigma \to \infty$:

$$\boxed{\mathrm{Bias}\big[\hat{\sigma}_{h_\sigma}(x)\big] \;=\; \tfrac{1}{2}\,h_\sigma^2\,\mu_2(K)\left[\sigma''(x) + 2\,\frac{\sigma'(x)\,p_X'(x)}{p_X(x)}\right] + o(h_\sigma^2) + O(r_0^{-1}).} \tag{12}$$

$$\boxed{\mathrm{Var}\big[\hat{\sigma}_{h_\sigma}(x)\big] \;=\; \frac{R(K)\,\sigma^2(x)\,(\kappa_4 - 1)}{4\,(r_0 - 1)\,n_0\,h_\sigma\,p_X(x)} \;+\; o\big((n_0 h_\sigma r_0)^{-1}\big).} \tag{13}$$

Two features distinguish (13) from a vanilla NW variance: (i) the $(r_0 - 1)^{-1}$ factor reflects the within-site replication, so doubling r_0 halves the variance contribution; (ii) the kurtosis factor $\kappa_4 - 1$ makes the variance sensitive to heavy-tailed noise (e.g. the Gamma or Student-t designs).

Optimal bandwidth and MSE rate. The pointwise MSE takes the canonical form $\mathrm{MSE}(x) = C_1(x)\, h_\sigma^4 + C_2(x)/(n_0 h_\sigma r_0)$, and minimising over h_σ yields

$$\boxed{h_\sigma^\star(x) \;\propto\; (n_0\, r_0)^{-1/5}, \qquad \mathrm{MSE}^\star(x) \;\propto\; (n_0\, r_0)^{-4/5}.} \tag{14}$$

Crucially, the *effective sample size* is $n_0 r_0$, not just n_0: each replicate sharpens $\tilde{\sigma}_i$ and propagates through the smoother. Thus under a fixed total budget $N = n_0 r_0$, the rate of $\hat{\sigma}$ depends only on N, while Stage-1 CKME fitting depends on n_0 alone — these two considerations pull (n_0, r_0) in opposite directions.

Propagating $\hat{\sigma}$ into the coverage bound. Plugging (11) into the adaptive bandwidth rule $\hat{h}(x) = c\,\hat{\sigma}(x)$ and tracking the perturbation through the score density g in (8), a first-order expansion yields an updated form of the coverage bound of Proposition 9:

$$\big|\mathbb{P}(Y \in C(x) \mid X = x) - (1-\alpha)\big| \;\le\; \underbrace{\tfrac{\pi^2 c^2}{6}\,\|g''\|_1}_{\text{smoothing}} + \underbrace{\|\widehat{F} - F\|_\infty}_{\text{CKME fit}} + \underbrace{c\,\|g'\|_\infty\,(n_0 r_0)^{-2/5}}_{\text{plug-in }\hat{\sigma}} + p_{n+1}. \tag{15}$$

The plug-in term inherits the MSE $^{1/2}$ rate $(n_0 r_0)^{-2/5}$ from (14) and enters linearly in c: larger c not only amplifies the smoothing bias but also the plug-in bias.

Implication: the optimal c shrinks under plug-in. Let $c^\star_{\mathrm{or}}$ minimise the oracle bound $\frac{\pi^2 c^2}{6}\|g''\|_1 + \|\widehat{F} - F\|_\infty$ and $c^\star_{\mathrm{pi}}$ minimise (15). Ignoring the constant CKME term, differentiating in c gives

$$c^\star_{\mathrm{pi}} \;\approx\; c^\star_{\mathrm{or}} \;-\; \frac{3\,\|g'\|_\infty}{\pi^2\,\|g''\|_1}\,(n_0 r_0)^{-2/5},$$

so the plug-in optimum is *smaller* than the oracle one, with a gap that closes at rate $(n_0 r_0)^{-2/5}$. This matches the empirical observation in the plug-in variant of the homogeneity experiment, where the best c on the c-scan shifts leftwards relative to the oracle curve, and the gap narrows as we enlarge r_0.

3.2 Bandwidth Rate for the CKME Indicator

Proposition 9 bounds the coverage deviation by $\frac{\pi^2 c^2}{6}\|g''\|_1 + \|\widehat{F} - F\|_\infty$, where the first term is a *bias* contribution from $h(x) = c\,\sigma(x)$ smoothing the score density g, while the second bundles the entire CKME estimation error. Hidden inside $\|\widehat{F} - F\|_\infty$ is a second, symmetric bias–variance trade-off in the bandwidth h itself, which we unpack here.

Bias and variance of $\widehat{F}(y|x)$. With a smooth indicator $\psi_h(y-t) \approx \mathbf{1}\{y \le t\}$ of bandwidth h, the CKME CDF estimator $\widehat{F}(y|x) = \sum_i c_i(x)\,\psi_h(y - Y_i)$ admits the pointwise decomposition

$$\boxed{\mathbb{E}[\widehat{F}(y|x)] - F(y|x) \;=\; \tfrac{1}{2}\,h^2\,\mu_2(\psi)\,f'_{Y|X}(y|x) \;+\; \underbrace{O\big(\ell_x^2\big)}_{\text{NW bias in } x} \;+\; o(h^2),} \tag{16}$$

$$\boxed{\mathrm{Var}[\widehat{F}(y|x)] \;=\; \frac{F(y|x)(1 - F(y|x))}{n_0\, r_0\, p_X(x)\,\ell_x} \;+\; O\big(h^{-1}/(n_0 r_0)\big) \;+\; o(\cdot).} \tag{17}$$

Here $\mu_2(\psi) = \int u^2 \psi'(u)\,du$ plays the role of the second moment of the induced kernel $\psi'_h = h^{-1}\psi'(\cdot/h)$. Equation (16) says h acts on the CDF exactly as a kernel smoother on F, and its bias is driven by the *score density slope* $f'_{Y|X}(y|x) = \partial_y f_{Y|X}(y|x)$.

Rate for $h^\star$. Balancing the h^4 bias^2 in (16) against the $h^{-1}/(n_0 r_0)$ variance contribution gives the classical Nadaraya–Watson rate

$$\boxed{h^\star \;\propto\; (n_0\, r_0)^{-1/5}, \qquad \big\|\widehat{F} - F\big\|_\infty^\star \;\propto\; (n_0\, r_0)^{-2/5}.} \tag{18}$$

Note the *same* effective sample size $n_0 r_0$ as for $\hat{\sigma}$ in (14): replications help $\widehat{F}$ too, because each $Y_{i,j}$ contributes an independent vote to the indicator average. This is why the replicated two-stage design (n_0, r_0) with $r_0 \geq 2$ dominates a single-shot design of the same total $N = n_0 r_0$ in the tail-quantile regime (where variance dominates).

Combining with the coverage bound. Substituting (18) into the coverage decomposition and collecting terms, Proposition 9 sharpens to

$$\big|\mathbb{P}(Y \in C(x) \mid X = x) - (1-\alpha)\big| \;\lesssim\; \underbrace{\tfrac{\pi^2 c^2}{6}\|g''\|_1}_{c\text{-smoothing}} + \underbrace{(n_0 r_0)^{-2/5}}_{h\text{ trade-off}} + \underbrace{c\,(n_0 r_0)^{-2/5}\|g'\|_\infty}_{\hat{\sigma}\text{ plug-in}} + p_{n+1}. \tag{19}$$

Three distinct bias–variance mechanisms are visible: (i) the homogeneity term driven by c, which *cannot* be driven to zero at finite n because $c \to 0$ sends the adaptive mechanism off; (ii) the CKME bandwidth h term, which vanishes at rate $(n_0 r_0)^{-2/5}$; and (iii) the plug-in $\hat{\sigma}$ term, also at $(n_0 r_0)^{-2/5}$ but scaled by c.

Remark 11 (Why high-τ quantiles do not converge at fixed h). Inverting (16) gives $\hat{q}_\tau(x) - q_\tau(x) \approx -\frac{1}{2}h^2\mu_2(\psi)\, f'_{Y|X}(q_\tau|x)/f_{Y|X}(q_\tau|x)$, so the quantile bias is amplified by the reciprocal tail density $1/f_{Y|X}(q_\tau|x)$. When $\tau \to 1$ this factor diverges for any distribution with a density tail, so *fixing* h (no matter how small) leaves a floor in $\sup_x |\hat{q}_\tau - q_\tau|$ that does not vanish as $n \to \infty$. Only $h \to 0$ at the rate prescribed by (18) restores convergence, and even then only if $h^2/f_{Y|X}(q_\tau|x) \to 0$ — a τ-dependent condition. This matches the empirical finding in `exp_onesided/exp2_sup_vs_tau.py`, where $\sup_x |\hat{q}_\tau - q_\tau|$ decays with n for $\tau \in \{0.5, 0.6, 0.7\}$ but stalls for $\tau \in \{0.8, 0.9, 0.95\}$ under the project's default fixed h.

Summary: three U-shapes, one budget. Equation (19) predicts *three* bias–variance U-shapes along the CKME pipeline: one in the CKME bandwidth h, one in the adaptive coefficient c, and one in the plug-in smoother bandwidth h_σ. All three are confirmed empirically in Figure 1: panel (a) varies h at fixed $(c, \hat{\sigma})$ and shows quantile error as a function of h across multiple τ levels; panel (b) varies c in $h(x) = c\,\hat{\sigma}(x)$ and reports the resulting quantile and coverage errors; panel (c) varies the plug-in smoother bandwidth h_σ and reports RMSE of $\hat{\sigma}$, KS-max of the score, and the sup coverage gap. All three share the effective sample size $n_0 r_0$, so budget allocation — not just parameter tuning — becomes the correct lens: the marginal return of one more replication equals that of one more site whenever variance dominates at that position on the U-curve.

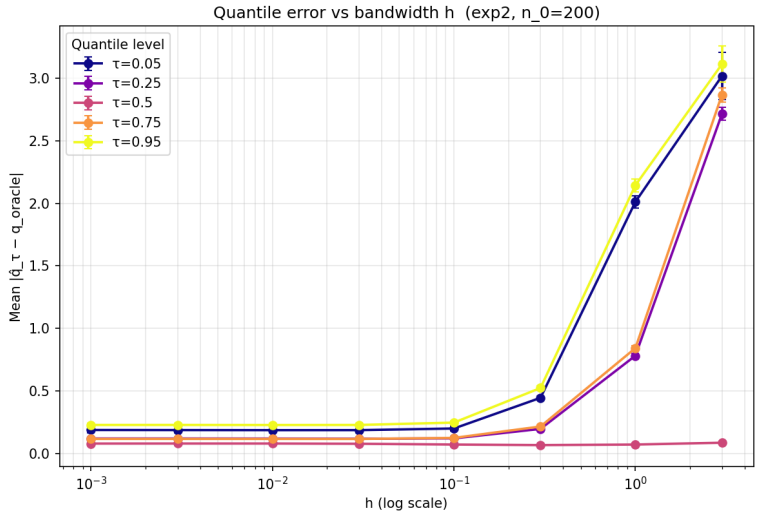
3.3 Generalization Beyond Gaussian Noise

The effective-bandwidth argument (Section 2.1) uses only the location-scale structure $Y = f(x) + \sigma(x)\varepsilon$, not the specific form of G. The key algebraic step

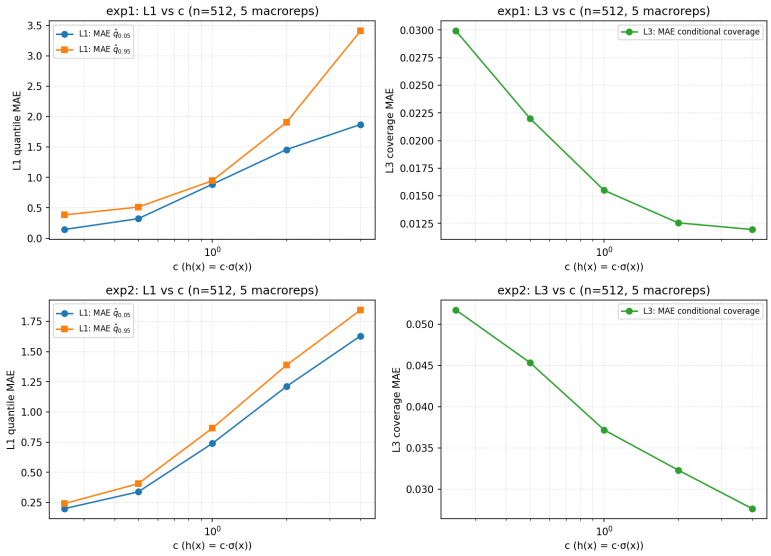
$$\frac{y - Y_{ij}}{h(x)} = \frac{z - Z_{ij}}{c}$$

holds regardless of whether ε is Gaussian, Student-t, centered Gamma, or any other distribution with $E[\varepsilon] = 0$, $\mathrm{Var}(\varepsilon) = 1$.

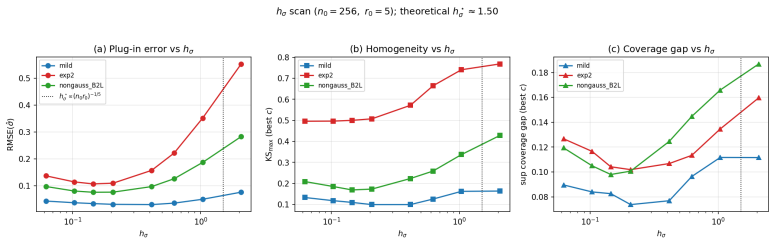
Therefore:



(a) CKME indicator bandwidth h (exp2): quantile error $|\hat{q}_\tau - q_\tau|$ vs. h across τ.



(b) Adaptive coefficient c: quantile and coverage errors vs. c in $h(x) = c\,\hat{\sigma}(x)$.



(c) Plug-in smoother bandwidth h_σ: RMSE of $\hat{\sigma}$, KS-max, and sup coverage gap vs. h_σ for three simulators.

Figure 1: **Three bias–variance U-shapes along the CKME pipeline.** Each panel isolates one bandwidth while holding the others fixed. The left branch of every U is variance-dominated and the right branch is bias-dominated, consistent with Equations (16)–(17), (19) and (12)–(13).

- Proposition 2 applies to all location-scale families.
- The proof requires only that G has a bounded density (for kernel smoothing consistency), not that G is Gaussian.
- Beyond location-scale (e.g., shape parameter varying with x), the standardization argument breaks. However, experiments show approximate homogeneity is preserved for moderate violations (Section 4).

4 Empirical Validation

4.1 Objective

The experiment tests three theoretical predictions:

1. Adaptive $h(x) = c\,\sigma(x)$ produces lower KS distance between local and pooled score distributions (score homogeneity).
2. Score homogeneity, not estimation accuracy, determines conditional coverage gap (decoupling).
3. The mechanism holds for non-Gaussian location-scale families (generality).

4.2 Three-Phase Design

We use three simulators with increasing difficulty, all sharing the true function $f(x) = e^{x/10}\sin(x)$ on $x \in [0, 2\pi]$:

Phase	**Simulator**	$\sigma(x)$ **profile**	**Purpose**
1	`mild`	$0.3(1 + 0.5\sin x)$, ratio $\approx 3\times$	Clean theory verification
2	`exp2`	$\sqrt{0.01 + 0.2(x-\pi)^2}$, ratio $\approx 200\times$	Finite-sample stress test
3	`nongauss_B2L`	$\sigma_{\text{tar}}(x) = 0.1 + 0.1(x-\pi)^2$, Gamma($k$=2)	Non-Gaussian robustness

Why three phases? Phase 1 (mild) has a small σ-ratio, so finite-sample noise is small and the effective-bandwidth mechanism can be verified cleanly. Phase 2 (exp2) has a σ-ratio of $\sim$200$\times$, stressing the mechanism where $\sigma(x) \to 0$ creates near-degenerate CDF estimates. Phase 3 (nongauss_B2L) uses centered Gamma noise ($k = 2$, strong skew), testing whether the location-scale argument extends to non-Gaussian settings.

4.3 Bandwidth Configurations

Nine bandwidth configurations per simulator:

- **3 fixed:** $h = 0.05$ (small), h_{cv} (cross-validated), $h = 0.5$ (large).
- **6 adaptive:** $h(x) = c\,\sigma(x)$ with oracle $\sigma(x)$, $c \in \{0.5, 1.0, 1.5, 2.0, 3.0, 5.0\}$.

All configurations share the same $(e\ell_x, \lambda)$ hyperparameters (cross-validated jointly), isolating the bandwidth effect.

4.4 Budget and Evaluation

- **Stage 1:** $n_0 = 512$ sites, $r_0 = 1$ replication each. Stage 2: $n_1 = 512$, $r_1 = 1$.
- **Evaluation grid:** $M = 50$ equally-spaced points in $[0, 2\pi]$.
- **MC test draws:** $B = 2000$ fresh draws per eval point per macrorep for conditional coverage and score ECDF.
- **Macroreplications:** $n_{\text{macro}} = 20$ (5 completed for current results).

4.5 Metrics

1. **KS**$_{\text{max}}$: $\max_{m=1}^{M} \text{KS}(F_{s|x_m}, F_{s,\text{pooled}})$ — score homogeneity (lower = better).
2. **KS**$_{\text{mean}}$: $\frac{1}{M}\sum_m \text{KS}(F_{s|x_m}, F_{s,\text{pooled}})$ — average homogeneity.
3. **Coverage gap (sup)**: $\max_m |P(Y \in C(x_m)|x_m) - (1-\alpha)|$ — worst-case conditional coverage deviation (lower = better).
4. **Coverage gap (MAE)**: $\frac{1}{M}\sum_m |P(Y \in C(x_m)|x_m) - (1-\alpha)|$ — average coverage deviation.
5. **Quantile error (sup)**: $\max_m \max_{\tau\in\{\alpha/2,1-\alpha/2\}} |\hat{q}_\tau(x_m) - q_\tau(x_m)|$ — estimation accuracy (not directly related to coverage under homogeneity).
6. **Width std**: $\text{SD}_x(W(x_m))$ — spatial variation of interval width (higher = more locally adaptive).

5 Discussion

5.1 Theory–Experiment Alignment

The experiments provide consistent support for all three theoretical predictions (based on $n_{\text{macro}} = 5$; further macroreplications are planned):

1. **Proposition 2:** Adaptive h reduces KS_{max} by 2.6×–3× vs. best fixed (Phase 1 and 3). Phase 2's smaller improvement (1.7×) reflects the finite-sample limitation where $\sigma(x) \to 0$.
2. **Decoupling (Section 2.4):** The anti-diagonal pattern in the decoupling plots — estimation accuracy vs. coverage gap — is consistently observed across all three phases. Fixed bandwidths cluster in the "accurate but inhomogeneous" corner; adaptive bandwidths move toward the "less accurate but homogeneous" corner with better coverage.
3. **Location-scale generality (Section 3.3):** Phase 3 (Gamma noise) shows the same qualitative pattern as Phase 1 (Gaussian noise), confirming that the mechanism depends on the location-scale structure, not the specific noise distribution.

5.2 Finite-Sample Limitations

Phase 2 (`exp2`) reveals important limits:

- When $\sigma(x) \to 0$ (at $x \approx \pi$), even adaptive $h(x) = c\,\sigma(x) \to 0$ creates a near-degenerate CDF with $O(h(x))$ resolution, and the KS distance from the pooled distribution remains high.

- Very large c ($c = 5.0$) achieves the best $\mathrm{KS}_{\max}$ but the worst coverage gap (sup), illustrating the transition toward the over-smoothed collapse regime (Proposition 4).
- The c-scan Pareto plot reveals an optimal range $c \approx 1$–2 that balances homogeneity and local adaptivity.

5.3 Practical Implications

1. **Bandwidth selection:** Based on the current three-phase evidence, $c \approx 1.5$ appears to be a reasonable default. The coverage gap is relatively insensitive to c in the range $[1, 2]$ across all three phases.
2. **Plug-in estimation:** In practice, $\sigma(x)$ is unknown. The two-stage design provides replication-based estimates $\hat{\sigma}(x_j)$ from Stage 1, enabling $\hat{h}(x) = c\,\hat{\sigma}(x)$. The simulation evidence motivates this plug-in approach, but quantifying how $\hat{\sigma}(x)$ estimation error degrades the homogeneity bound is addressed by the rate analysis in Section 3.1.
3. **CDF-first advantage:** The score homogeneity mechanism is specific to CDF-based scores (Remark 3). Quantile-first methods (CQR, DCP-QR) cannot exploit this regime, even with adaptive bandwidth.

5.4 Comparison with Related Theoretical Frameworks

We position our score homogeneity results relative to two recent frameworks for conditional conformal prediction: CP^2 [1] and RCP [2]. Table 1 summarizes the comparison.

vs. CP^2 (Gibbs, Cherian, Candès 2023). CP^2 provides a general conditional coverage framework via RKHS regularization. It bounds the conditional coverage gap by $\|f\|_{\mathcal{H}} \cdot R_n$, where $\|f\|_{\mathcal{H}}$ is a score distribution regularity term and R_n a calibration error. This is a powerful "black-box" result: it applies broadly but does not identify *when* the score distribution is well-behaved, nor does it connect to any specific method parameter.

Our contribution fills this gap in two ways:

1. **Sufficient condition.** We identify a concrete, verifiable condition — $h(x) = c\,\sigma(x)$ under a location-scale model — that makes the CDF-based score distribution x-independent.
2. **Explicit quantitative bound.** Using the classical result of Devroye and Györfi (1985) [4] for the logistic kernel, we obtain

$$d_{\mathrm{TV}}(\psi_c * g,\ g) \leq \frac{\pi^2 c^2}{6}\,\|g''\|_1,$$

where ψ_c is the logistic kernel scaled by c. This yields $d_{\mathrm{TV}}(F_{s|x}, F_{s,\mathrm{pivot}}) \leq \frac{\pi^2 c^2}{6}\,\|g''\|_1 + O_p(n^{-s})$, separating smoothing bias from regression error. The bound is computable: given c and the noise density g, one evaluates $\|g''\|_1$ directly.

In short, CP^2 says "if the score distribution is regular enough, coverage is good"; we say "under adaptive bandwidth, the score distribution *is* regular, and here is the explicit bound."

vs. RCP (Rectified Conformal Prediction). RCP addresses the same problem — score heterogeneity degrades conditional coverage — but operates on a fundamentally different error decomposition (see Section 3 for the formal comparison). RCP's coverage bound is controlled by the *quantile-of-score estimation error* $\sup_x |\hat{\tau}(x) - \tau(x)|$, where $\hat{\tau}(x)$ estimates the conditional quantile of the rectified score. Our coverage is instead controlled by $d_{\mathrm{TV}}(F_{s|x}, F_{s,\mathrm{pivot}})$, which decomposes into smoothing bias $O(c^2)$ and CDF estimation error $O_p(n^{-s})$.

The practical consequence is that our approach achieves score homogeneity *without any post-hoc transformation*: under adaptive bandwidth, the CDF-first scores are structurally x-independent ("natural rectification"), whereas RCP requires (i) estimating a rectification map T, (ii) additional sample splitting, and (iii) Lipschitz regularity conditions on T. This is specific to CDF-based scores — quantile-first methods (CQR, DCP-QR) do not admit this mechanism.

Additional elements. Two aspects of our analysis, to our knowledge, are not addressed by either CP^2 or RCP:

1. **Two-condition framework (Section 2.4).** To our knowledge, neither CP^2 nor RCP explicitly distinguishes between "useful" and "trivial" score homogeneity. We show that $h \to \infty$ also produces x-independent scores (Proposition 4), but with degenerate constant-width intervals. Our joint requirement of homogeneity *plus* local adaptivity ($W(x)/\sigma(x) \to \mathrm{const}$) excludes this trivial regime and pins down the useful range $c \in [1, 2]$.

2. **Explicit bias–variance decomposition in c.** The total d_{TV} decomposes as $\underbrace{\frac{\pi^2 c^2}{6}\|g''\|_1}_{\text{smoothing bias}} + \underbrace{O_p(n^{-s})}_{\text{regression error}}$.

   Small c reduces smoothing bias but increases regression error (CDF estimate variance); large c does the reverse. This decomposition provides an explicit, operational guide for bandwidth selection in the conformal prediction setting, and it is consistent with the experimentally observed optimal range.

Table 1: Comparison of conditional coverage frameworks.

	CP^2	RCP	Ours
Sufficient condition for homogeneity	$\times$	via $T(s,x)$	$h = c\,\sigma(x)$
Quantitative bound	implicit (RKHS)	Lipschitz	$\pi^2 c^2/6 \cdot \|g''\|_1$
Extra estimation step	—	estimate T	none
Excludes over-smoothed collapse	$\times$	$\times$	✓
Bandwidth selection guidance	$\times$	$\times$	✓
Scope	general	general	location-scale

Scope trade-off. Our results apply to the location-scale family $Y = f(x) + \sigma(x)\varepsilon$, which is narrower than the settings of CP^2 or RCP. Within this class, however, the analysis yields more explicit and operational conclusions: a verifiable sufficient condition, a computable bound, and a principled bandwidth selection rule. For applications where location-scale is a reasonable model (simulation, engineering reliability, heteroscedastic regression), these concrete results complement the broader but more implicit guarantees of CP^2 and RCP.

References

[1] Gibbs, I., Cherian, J.J., and Candès, E.J. (2023). Conformal prediction with conditional guarantees. *arXiv:2305.12616*.

[2] Bhatnagar, A., Wang, H., Xiong, C., and Bai, Y. (2025). Improved conditional coverage via rectified conformity scores. *arXiv:2502.16336v2*.

[3] Barber, R.F., Candès, E.J., Ramdas, A., and Tibshirani, R.J. (2023). Conformal prediction beyond exchangeability. *Annals of Statistics*, 51(2), 816–845.

[4] Devroye, L. and Györfi, L. (1985). *Nonparametric Density Estimation: The L_1 View*. Wiley, New York.